One argument, several ways to read it

An architectural companion to Lean Exposition

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A mathematician reading a short proof may want to stop at its main idea, check one difficult estimate, or learn the subject from the beginning. Those are different requests. A proof assistant can check a precisely stated theorem, but its successful check does not choose an explanation for any of these readers. Nor does a readable explanation, by itself, show how faithfully it represents the checked statement.

Lean Exposition connects three written accounts of selected mathematics to a shared record of claims, assumptions, sources and checks. Each account has its own purpose and reading order. Their connections let a reader leave an explanation at the point of doubt, recover the relevant detail, and return without losing track of which statement was being discussed. This companion explains that arrangement and the choices behind it. The mathematical papers supply the arguments themselves.

1. Why write more than one account?

The starting material may include an original research paper, a formal development in the Lean proof assistant, and earlier literature. These sources answer different questions. The research paper explains a mathematical argument in the author's chosen language. The Lean development gives precise definitions and formal proofs. Earlier literature establishes what was known, where an input came from and which conclusions require additional results. Reading one does not necessarily recover the content of the others.

An author first reconstructs the selected argument in a **long reasoning record**. This is where a reader should be able to find the derivation behind an estimate, the treatment of an endpoint, or the reason two formulations correspond. Its length follows the argument that needs recovering. It can retain a failed approach when the failure explains why a hypothesis matters. It also identifies steps that have not been justified.

A **short research paper** then selects a contribution worth presenting as a connected argument. It gives the statement, its position in the literature and the decisive proof, with precise destinations for substantial details it omits. The selection may leave whole further results in the long record. Those omissions should make the paper more focused without making its claims stronger.

An **expository paper** makes a different selection. It teaches a result to readers with stated prerequisites, introducing objects through examples and explaining why each construction is needed. It may spend more time on a definition than the specialist paper, while covering fewer consequences.

Its author keeps the original sources and detailed record available; the exposition need not be produced by repeatedly shortening the short paper.

The supplied example concerns the Sherrington–Kirkpatrick (SK) model, a random optimisation problem on interacting spins. Its exposition begins with a two-spin calculation and explains why an optimal function used to express the model's value cannot stay flat on an interval. The short paper presents this no-gap theorem and related consequences for a reader familiar with stochastic calculus. The long record reconstructs the analytic steps and continues into the separate argument about the Quantum Approximate Optimization Algorithm (QAOA). A reader interested in the no-gap mechanism does not have to read the circuit construction first.

These are entrances for different purposes. The writing process gives authors more context than any one reader is asked to absorb.

2. What the shared graph remembers

Maintaining three accounts creates a practical problem. If a hypothesis is corrected in the detailed argument, which shorter passages need attention? If a reader doubts an estimate, where is it explained, and which source is being invoked? A bibliography alone cannot answer these questions at the level of a particular passage.

The project records selected claims and their relationships in a graph. A claim has a stated scope and associated assumptions. It can point to an ingredient used in its argument, an original source, or a formal statement with which the author proposes a correspondence. Sources retain their editions; checking records say what was checked and with what outcome. This is a curated, partial account of the available mathematics. It does not pretend to contain every declaration or every dependency in the underlying Lean development.

Each written paper has a **passage map**: annotations connecting selected passages to the claims, assumptions and citations they discuss. Thus two papers can refer to the same claim while explaining it differently. Their prose remains independently readable, and text without a recorded mapping remains visible as such.

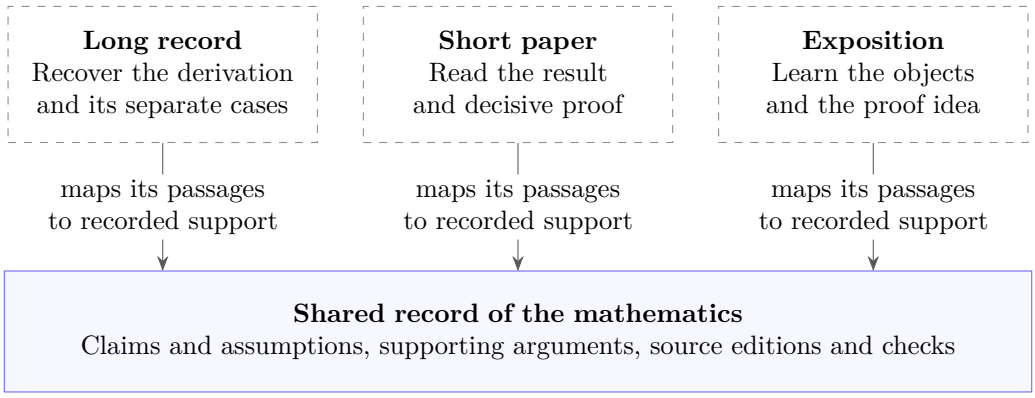


Figure 1: Why three papers need only one shared record. Each paper chooses its own reading order and connects its passages to relevant parts of the same corpus. The arrows denote authored correspondence, not automatic generation or proof. A paper can select fewer results while retaining the assumptions and sources its claims need.

An arrow needs a meaning. Here an argument arrow points from a claim to an ingredient it uses: following it asks what supports the claim. Following the same relation backwards asks what recorded arguments use that ingredient. The direction of the stored relationship stays the same. Several ingredients may be needed together; an arrow to one of them does not say that it alone proves the conclusion.

Other relationships answer other questions. A citation credits or refers to a source. A correspondence with a formal statement asks the reader to compare two formulations. A recorded Lean dependency says that a constant occurs in a declaration's formal type or proof, or in its structural references. A checking record identifies the source or claim it concerns. These relations must remain distinguishable: a source can be historically relevant without being a premise, and a proposed correspondence can require a mathematical argument of its own.

The order of paragraphs is another relationship again. An exposition may give an example before stating the general theorem; a specialist paper may state the theorem first. Neither order changes the theorem's prerequisites. A paper's reading route therefore consists of its authored sequence together with the support selected for its passages. The graph helps readers inspect that selection; it does not choose a sufficient proof or a good teaching order.

3. Follow one claim back to its argument

The SK model asks which choices of signs maximise a random sum of pair interactions. The Parisi formula expresses its limiting optimum through a minimisation over functions called order parameters. In the pure model at zero external field, an admissible order parameter is a nonnegative, nondecreasing, right-continuous, integrable function $\gamma : [0,1) \to [0,\infty)$. Its associated measure records the increments of γ. A flat interval is a gap in those increments, even if the function's value there is positive.

The result at issue says that an admissible minimiser has no such gap: its measure has full support relative to $[0,1)$. The endpoint one is excluded. Existence of a minimiser comes from a separate theorem of Auffinger and Chen. An earlier paper by Hong-Bin Chen established full support by another method. The worked accounts reconstruct the quantitative argument in OpenAI's full-support manuscript. Their contribution here is the explanation and its correspondence to sources.

Consider just one case: a bounded gap (ℓ, r) between support points, with $0 < \ell < r < 1$, on which γ has a positive constant value. To test whether γ could minimise, the proof constructs a differentiable curve q measuring the average squared response in an auxiliary optimisation problem. The mathematical papers explain that construction. Here we take two properties as inputs and follow how they exclude the gap.

Minimality gives **contact** at the endpoints. In the notation of the papers, this supplies

$$q(\ell) = \ell, \qquad q(r) = r, \qquad q'(\ell), q'(r) \le 1.$$

The change in q across the interval must therefore be $r - \ell$. Separately, a strict-curvature estimate on a positive constant interval makes q' strictly convex. Since q' is continuous at the endpoints, its interior values lie strictly below their joining chord, which is at most one. Consequently

$$q(r) - q(\ell) = \int_\ell^r q'(t)\,dt < r - \ell.$$

The two requirements contradict each other. Neither endpoint contact nor strict curvature alone gives this contradiction.

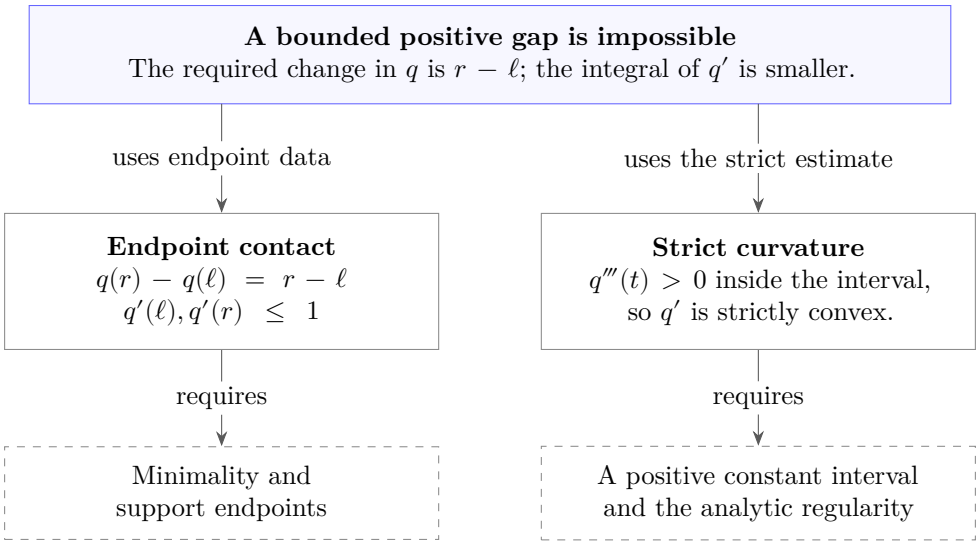


Figure 2: Which ingredients does the contradiction use? Arrows point from a conclusion to its inputs. Both branches are needed in the stated SK setting. Minimality supplies contact; the curvature estimate itself applies without minimality. This is one branch of the ordinary argument, not an extracted Lean proof graph. Zero coefficients and initial or terminal gaps require separate treatment.

Now suppose a reader accepts the final integral comparison but wants to know why the curvature is strict. The short paper, section "Minimality gives contact; curvature excludes gaps" identifies the estimate at its use. The long record, section "Strict curvature on a constant interval" reconstructs the analytic proof, including the integrated inequality and the passage from simpler coefficients to the stated class. It leads in turn to Proposition 5.1 of the original manuscript. Having checked that destination, the reader returns to the short proof and uses the estimate together with contact.

A reader who does not yet understand a gap can instead enter the exposition, section "What a gap means". Its example separates a positive cumulative value from vanishing increments. The later section "Why a flat interval cannot minimise" develops the contact and area argument. This route teaches the objects that the specialist route assumes. The graph connects the example to the Stieltjes-measure definition and records its use in the later argument; reading each passage supplies the explanation.

4. What survives a shorter account

Shortening the example cannot change "an admissible minimiser" into "an admissible function". Constant functions are allowed candidates, and the curvature estimate does not say they minimise the functional. Without minimality, the endpoint contact used in the contradiction is missing. The short account must retain that premise where it states and uses the no-gap conclusion.

It must also preserve the result's extent. Full support says that every nonempty relatively open interval receives mass. It does not alone assert a density, let alone a strictly positive density. The stronger density conclusion in the source has a further argument. Likewise, the broader QAOA claim requires approximation and circuit arguments beyond full support. Different accounts may stop at different conclusions, provided each makes its own stopping point clear.

The passage maps help enforce this discipline. For the claims an author selects, the checks require the recorded assumptions, citations and unresolved obligations to remain in the map. They can catch a missing reference or an unreconciled change to mapped text. They cannot tell whether the theorem's wording still states the assumption, or whether the cited argument is adequate. Those are questions for mathematical and editorial review.

Attribution survives the same selection. The no-gap theorem retains credit for the earlier support result and for the quantitative proof being explained. A reference may supply a theorem, an adapted construction or historical context. Recording that role avoids treating every citation as a logical dependency while still letting the reader recover the relevant source.

5. Explanation, formal evidence and navigation

A reader may next ask whether the no-gap conclusion has a formal counterpart. The project records a correspondence with a selected Lean statement. That statement assumes an admissible minimiser on a Brownian system and supplies full support, strict increase and associated moment identities. It does not assert minimiser existence or QAOA circuit convergence. Its original formal statement is a destination for comparison, rather than a formal certificate for every sentence of the exposition.

Even a close correspondence may need explanation. The long record's section "The support statement and the derivative bridge" compares a formal limit along prescribed step sizes with the ordinary right derivative used in the manuscript. A convexity argument connects them. Following a link finds that argument; it does not perform it.

The design therefore preserves three distinct activities. The written account explains why a conclusion follows. Formal checking tests a precise statement and proof in a specified environment. Navigation helps a reader find the account, formal statement and records of the checks. None can stand in for the other two.

Checking outcomes also retain their scope. Identifying unchanged source bytes, building a Lean project, comparing it with a separately stated challenge, and reviewing its mathematical interpretation answer different questions. For the retained SK research example, the corrected formal comparison and dependency extraction have no recovered final outcome. Their results remain unknown, and this companion establishes no new formal acceptance. Earlier unsuccessful runs remain observations of those runs, rather than counterexamples to the theorem.

Reviews concern particular editions. If an author changes a hypothesis or maps the same paragraph to a different formal statement, the correspondence has changed even when much of the prose has not. The project keeps the text, map and sources identifiable together so a review can be understood in its original context. A revised edition needs review of the affected claim.

6. What a reader or author can use today

Today the repository provides the written drafts, their source maps and local tools for inspecting them. A reader can begin with a paper passage and find its recorded assumptions and support, look for other accounts of the same claim, or begin with a source and find its recorded uses. The tools can follow selected kinds of relationship forwards or backwards. They retain the links among the records found, rather than replacing a shared argument by a single chain.

Each query has a boundary. It may be restricted to one paper or stop after a chosen amount of exploration. The returned boundary distinguishes such limits from the absence of another recorded

connection. Because the corpus is partial, finding no connection cannot establish mathematical independence or a complete account of a source.

The papers and their maps can also be packaged together for inspection in another application. That package contains the captured writing and records; some original sources remain external. Source references identify what should be consulted, but cannot guarantee that an unavailable source will later be retrievable. The current tools provide local queries and a programming interface. The repository also includes a reading website that consumes a captured edition: readers can unfold the argument, follow a passage to its recorded claims, and move between the three accounts. The website displays the same source and checking records; it does not supply additional proof or turn an authored correspondence into a verified one.

Review asks two separate questions: does the writing faithfully represent the mathematics, and can the intended reader follow it? A useful reader task for this example is to explain why contact and curvature conflict, then say which step fails without minimality. Finding the right source is helpful, but does not by itself answer that task. The drafts have received source-based agent review; independent human acceptance and evidence of improved comprehension remain to be obtained.

7. How the writing methods improve

The writing process is organised into four **authoring transitions**. An author first records the sources and their mathematical support, then reconstructs the selected argument, selects the expert account and develops an exposition for a stated reader. Each transition has a **transition skill**: a written method describing the material to inspect, decisions to make and questions to check. These methods guide an author or an assistant. They are not programs that generate a proof or certify an explanation.

The inputs accumulate through the funnel; the scope of each paper is chosen afresh. An exposition can therefore teach one result using the full argument without inheriting every topic or the expert paper's order. A correction can also travel back: a question from an elementary reader may expose a missing assumption in the detailed argument. The claim must then be repaired wherever it is used, rather than only in the passage where the difficulty was noticed.

Review improves two different things. First, it can repair the present account. Second, it can reveal a lesson worth retaining in the method for another problem. The **review-and-refinement loop** keeps those outcomes distinct. An author preserves the reviewed edition, tests the criticism against the sources, revises the affected passages and checks their uses in the other accounts. The author then asks whether the failure came from missing guidance, guidance that was not followed, or a choice particular to this paper. Only the first needs a new instruction; the second needs its existing instruction to reach the decision, and the third may remain a local choice.

For example, the SK exposition uses a finite calculation with accumulated weights to prepare an integral and later returns to the same accounting in the contact argument. The transferable lesson is not that every exposition needs those weights. It is to let the reader perform an unfamiliar operation before relying on it in a proof, then use the example again when its purpose becomes clear. A specialist who already knows the operation may need no such preparation. A further question changes a hypothesis enough to change the answer: carrying out the same calculation does not by itself show when the argument applies. Before retaining a lesson as general guidance, the author tests it on differently shaped problems and a case where it should not apply. Missing mathematical support returns to research, however clear the examples become.

The accepted lesson belongs in the transition skill that owns the decision, or in the shared authoring guide when several stages need it. The record of what happened stays with the reviewed edition. This lets the next author use the method without reading every earlier review, while preserving the evidence behind a change. The complete working instructions are kept in this repository; their historical exemplars are optional reading. Checks can establish that text and references still agree with their recorded editions. They cannot establish that an author applied a skill well or that a reader learned more.

8. Why keep this much structure?

Ordinary papers with careful references remain useful. The additional work here is maintaining explicit relationships across several accounts, so that a reader or an author can find a claim's assumptions, locate a missing derivation and identify recorded passages affected by a correction. The structure earns its place only if these journeys are useful. It also costs time: maps need authoring, changes need reconciliation, and correspondence still needs mathematical judgment.

The arrangement builds on existing practices. In particular, leanblueprint already connects informal statements with formal declarations and their dependencies. Lean Exposition's emphasis is the maintained relationship between several authored accounts, their selected passages and the distinct evidence behind them. It makes no claim to have invented informal-to-formal links or demonstrated a comparative learning benefit.

One shared corpus avoids maintaining a separate version of each claim for every paper. Separate annotations allow each account to read as a paper, while the repository's guides carry the operating instructions. Readers can therefore understand the design before deciding whether to use the software. The intended result is a shorter route from "I do not understand this step" to the argument that resolves it, with its assumptions and credit still attached.

Figure credit. The diagrams adapt the shared-support and parallel-record compositions previously used in this companion, derived from Will Cook's *A Repository-Based System for Research and Publication* (CC BY 4.0). Their subjects, labels and reading order have been revised for this account.